

HOMER, AK, USA

WMO: 703410

Lat: 59.642N

Lon: 151.491W

Elev: 20

StdP: 101.09

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25507

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-15.0	-23.5	0.5	-14.5	-21.4	0.6	-13.2	12.1	0.3	10.9	-0.8	2.3	40	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	18.9	13.9	17.7	13.3	16.7	12.7	14.6	18.0	13.9	16.9	13.3	16.0	3.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	9.2	15.9	12.3	8.9	15.2	11.6	8.5	14.6	40.7	18.1	38.8	17.0	37.2	16.1	19.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.4	7.5	DB	-19.4	22.2	3.1	2.0	-21.6	23.7	-23.5	24.8	-25.2	26.0	-27.5	27.4	(4)
(5)			WB	-19.8	15.9	3.0	1.2	-22.0	16.8	-23.7	17.5	-25.4	18.1	-27.5	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-3.8	-2.1	-1.3	3.4	7.4	10.8	12.9	12.5	9.5	4.5	-0.8	-2.9	(6)
(7)		DBStd	7.19	6.22	5.12	4.54	2.87	2.47	2.06	1.82	1.93	2.42	3.60	5.00	5.29	(7)
(8)		HDD10.0	2353	427	339	352	199	87	13	1	2	37	173	324	399	(8)
(9)		HDD18.3	5157	685	572	610	449	340	225	168	179	266	430	574	657	(9)
(10)		CDD10.0	238	0	0	0	0	6	38	92	81	21	2	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	2	0	0	0	0	0	1	1	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.8	2.9	3.0	3.1	3.0	2.7	2.5	2.6	2.7	2.8	3.0	(14)	
(15)	Precipitation	PrecAvg	645	64	51	43	29	27	24	38	60	81	78	68	78	(15)
(16)		PrecMax	968	220	143	153	104	59	63	95	133	157	217	222	199	(16)
(17)		PrecMin	364	10	3	1	2	2	1	5	12	21	21	4	3	(17)
(18)		PrecStd	135	47	32	33	20	16	14	22	27	36	45	53	50	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	8.2	9.3	12.9	17.5	20.4	21.4	20.6	17.3	13.5	10.7	8.5	(19)
(20)			MCWB	5.9	5.2	5.3	7.7	9.6	13.7	15.3	14.7	12.3	9.7	7.2	5.8	(20)
(21)		2%	DB	7.1	6.2	7.3	10.5	14.9	17.6	18.9	18.8	15.7	12.0	9.0	7.0	(21)
(22)			MCWB	4.6	3.9	4.5	6.0	9.2	12.4	14.2	14.0	11.5	8.5	6.4	4.6	(22)
(23)		5%	DB	5.5	4.9	5.9	9.2	13.1	16.2	17.7	17.4	14.4	10.8	7.5	5.8	(23)
(24)			MCWB	3.4	2.9	3.3	5.4	8.6	11.6	13.6	13.6	11.0	7.9	5.5	3.7	(24)
(25)		10%	DB	4.2	3.9	4.6	8.0	12.0	14.9	16.7	16.3	13.5	9.8	6.1	4.4	(25)
(26)			MCWB	2.7	2.1	2.4	4.7	8.0	11.0	13.1	13.0	10.6	7.3	4.5	2.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	5.6	5.9	8.3	10.9	14.5	15.6	15.7	13.2	10.4	7.9	6.2	(27)
(28)			MCDWB	9.1	7.7	8.5	12.5	15.2	19.2	20.2	19.1	15.9	12.8	9.8	8.2	(28)
(29)		2%	WB	4.7	4.2	4.8	6.6	9.8	13.1	14.7	14.7	12.3	9.1	6.7	4.8	(29)
(30)			MCDWB	6.8	6.0	7.0	9.6	13.8	16.7	18.2	17.8	14.5	11.3	8.6	6.7	(30)
(31)		5%	WB	3.6	3.1	3.7	5.7	9.0	12.1	14.0	14.0	11.6	8.3	5.7	3.9	(31)
(32)			MCDWB	5.3	4.5	5.6	8.6	12.7	15.3	17.1	16.7	13.7	10.3	7.4	5.5	(32)
(33)		10%	WB	2.7	2.3	2.6	5.0	8.2	11.3	13.4	13.3	11.0	7.7	4.6	2.9	(33)
(34)			MCDWB	4.1	3.7	4.3	7.8	11.6	14.3	16.2	15.8	12.9	9.6	6.1	4.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.0	6.2	6.7	7.1	7.4	7.0	6.4	6.9	6.4	6.1	5.8	(35)	
(36)			MCDBR	5.7	5.7	6.8	8.7	9.8	9.4	8.3	8.6	7.9	6.1	5.5	5.6	(36)
(37)			MCWBR	4.2	4.2	4.8	5.6	5.8	5.3	4.7	5.1	5.1	4.1	4.2	4.3	(37)
(38)		5% WB	MCDWB	5.5	5.4	6.0	8.0	9.2	8.6	7.6	7.7	6.8	5.5	5.1	5.3	(38)
(39)			MCWBR	4.3	4.2	4.4	5.3	5.6	5.1	4.4	4.8	4.7	3.9	3.9	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.302	0.309	0.352	0.370	0.358	0.364	0.356	0.333	0.314	0.272	0.258	(40)	
(41)		taud	2.207	2.263	2.335	2.283	2.309	2.428	2.454	2.480	2.514	2.409	2.166	2.005	(41)	
(42)		Ebn at Noon	550	704	829	847	856	875	860	840	803	691	495	347	(42)	
(43)		Edh at Noon	41	69	86	109	115	104	99	90	74	59	42	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.36	0.97	2.32	3.69	4.92	5.34	4.76	3.78	2.46	1.21	0.50	0.20	(44)	
(45)		RadStd	0.04	0.15	0.29	0.47	0.41	0.37	0.39	0.46	0.36	0.13	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+2.10	+1.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.74	+2.23	+2.07	N/A	N/A	N/A	N/A	-251	N/A	N/A

Nomenclature: See separate page